

Physics-guided discriminability of process-level observables in partially observable cancer systems

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Abstract

Bulk gene-expression profiles are widely used to construct pathway- and biological-process-level representations of cancer systems. However, bulk transcriptomic measurements are not direct readouts of unique biological states. They are projected observations shaped by cellular composition, biological organization, unresolved heterogeneity, and measurement variability. Under this partial-observability condition, scalar pathway activity scores may obscure differences in the internal structure and reliability of process-level observables. Here, we introduce a measurement-aware framework for evaluating process-level gene-expression observables by jointly considering activity magnitude and observable width. For each biological-process observable, patient-level activity was quantified by the mean expression of mapped genes, μ , while observable width was quantified by the full width at half maximum, W , of the within-process gene-expression distribution. In this formulation, μ represents the activity center of the projected process-level signal, whereas W provides an operational measure of observable cloud thickness. W is not treated as physical entropy or a direct measure of hidden-state complexity, but as a reproducible proxy for observable dispersion.

Across multiple cancer gene-expression cohorts, Hallmark and biological-process observables showed heterogeneous organization in joint μ - W space. Processes with comparable activity magnitude often exhibited different

observable widths, indicating that scalar activity alone does not fully describe process-level transcriptomic structure. Structured biological-process observables also differed from size-matched random gene sets, while high-discriminability random sets occasionally emerged, showing that clinical discriminability should be interpreted together with random-background behavior, width-aware coherence, and biological structure.

These results support a measurement-aware interpretation of pathway-level cancer transcriptomic observables. The main contribution is not that observable width universally improves discriminability, but that it changes how discriminability should be interpreted in partially observable cancer systems.

Keywords:

Partial observability; Gene expression; Biological processes; Hallmark pathways; Discriminability; Observable geometry; Uncertainty structure; Systems biology

1. Introduction

Cancer transcriptomes are commonly interpreted through molecular observables derived from bulk gene-expression measurements, including pathway scores, biological-process activities, and transcriptomic signatures. These representations are useful because they reduce high-dimensional expression profiles into biologically interpretable features for cancer classification, survival stratification, and mechanistic analysis. Hallmark signatures, Gene Ontology biological-process terms, Reactome pathways, and related gene-set-based approaches have therefore become widely used tools for summarizing coordinated transcriptomic programs in cancer systems [1–8].

However, bulk gene-expression-derived observables are not direct measurements of unique biological states. Tumor transcriptomic profiles are effective measurements generated from heterogeneous tissues containing

malignant cells, immune populations, stromal components, vascular structures, and other microenvironmental elements, together with biological fluctuation and measurement variability [9–17]. As a result, similar observed gene-expression patterns may arise from different underlying biological organizations. This limitation is particularly relevant in cancer, where intratumor heterogeneity, clonal evolution, tumor–microenvironment interactions, and therapy-associated adaptation strongly influence molecular behavior and clinical outcome [10–17].

Most pathway-level and biological-process-level analyses summarize grouped gene-expression measurements using scalar activity scores [5–8]. These approaches provide interpretable and practical coarse-grained representations, but they primarily emphasize activity magnitude. They do not explicitly describe the internal dispersion, compactness, or uncertainty structure of the observable itself. In practice, two biological processes may show comparable mean activity while differing substantially in the distribution of their member-gene expression values. Similarly, two patients may have similar pathway activity scores while the genes contributing to those scores show different degrees of coherence. Thus, scalar activity alone may be insufficient for evaluating process-level observables derived from heterogeneous bulk tumor measurements.

This limitation can be viewed as a partial-observability problem. Bulk transcriptomic profiles provide projected and incomplete molecular views of tumor organization rather than direct access to the full underlying biological state. In simplified form, an observed molecular signal can be written as

$$O = h(S) + \epsilon,$$

where S denotes the underlying biological organization of the tumor system, $h(\cdot)$ represents the observation process, O denotes the measured observable, and ϵ captures unresolved biological and measurement variability. If the observation mapping is non-injective, multiple latent biological configurations may generate similar observable patterns. Observable similarity therefore does not necessarily imply similarity of underlying biological organization or clinical behavior [18–26].

This perspective motivates a measurement-aware representation of process-level gene-expression observables. In physics, statistics, and complex-systems analysis, collective observables are often interpreted not only by their mean behavior but also by their fluctuation, dispersion, and distributional structure [27–33]. Motivated by this principle, we represent each biological-process observable using both its activity center and its observable width. For a biological process B in patient p , the patient-level observable is defined from the mapped gene-expression values,

$$O_{B,p} = \{Z_{g,p} : g \in B\},$$

where $Z_{g,p}$ denotes the cohort-wise z -score-normalized expression value of gene g in patient p . The activity center is quantified by the mean expression of mapped genes,

$$\mu_{B,p} = \frac{1}{N_B} \sum_{g \in B} Z_{g,p},$$

where N_B is the number of mapped genes in process B . The observable width is quantified by the full width at half maximum,

$$W_{B,p} = FWHM(O_{B,p}).$$

Here, $\mu_{B,p}$ represents the patient-level activity center of the projected process-level signal, whereas $W_{B,p}$ quantifies the within-process gene-expression width for that patient. $W_{B,p}$ is interpreted as an operational measure of observable cloud thickness and as a reproducible proxy for observable dispersion. It is not treated as literal thermodynamic entropy, a direct measure of hidden-state complexity, or the FWHM of pathway activity scores across patients.

Using this activity–width representation, we evaluated Hallmark and biological-process observables across multiple cancer gene-expression cohorts. We asked whether observable width and coherence provide information beyond scalar pathway activity, whether structured biological-process observables behave differently from size-matched random gene sets, and whether uncertainty-aware discriminability depends on cancer context and observation scale. The conceptual framework and analytical workflow are summarized in Figure 1, while

detailed preprocessing, gene-set mapping, normalization, FWHM estimation, survival analysis, random-baseline construction, and scale-sensitivity analyses are provided in the Supplementary Methods.

Across cancer cohorts, the analyses revealed heterogeneous organization of process-level observables in joint μ - W space, cancer-dependent responses to width-aware discriminability scoring, distinct but nontrivial behavior relative to size-matched random gene sets, and scale-dependent effects when varying the number of selected process-level observables (Figures 2–5; Supplementary Figures S1–S10). Together, these results support a measurement-aware interpretation of pathway-level cancer transcriptomic observables. The central contribution is not that observable width universally improves discriminability, but that it changes how discriminability should be interpreted. Process-level observables should therefore be evaluated not only by activity magnitude or survival association, but also by observable width, coherence, random-background behavior, biological structure, and observation scale.

2. Process-level observables as projected state clouds

We represented process-level gene-expression observables as distributional measurements rather than scalar activity scores alone. This representation allows each biological-process observable to be described by both an activity center and an observable width.

For an individual gene g , the observed expression signal can be written as

$$O_g^{obs} = h_g(S; C_{bio}) + \epsilon_g,$$

where S denotes the underlying biological organization of the tumor system, $h_g(\cdot)$ represents the gene-specific observation process, C_{bio} denotes biological context and constraints, and ϵ_g captures unresolved biological and measurement variability. In this formulation, O_g^{obs} is a projected observation of S , rather than a uniquely invertible readout of the tumor system. Similar observed

expression values may therefore arise from different underlying biological configurations [13–17,22–26].

This uncertainty should not be interpreted as technical noise alone. Bulk gene-expression signals may also reflect cellular heterogeneity, mixed tissue composition, pathway redundancy, compensatory signaling, regulatory coupling, and other biological sources of variation [10–17,35–38]. Single-gene measurements provide limited access to this structure because each gene is observed as an isolated scalar quantity. By contrast, biological-process and Hallmark gene sets provide collective observable ensembles that group coordinated gene-expression signals into process-level representations [1–6]. These collective observables do not eliminate projection uncertainty, but they provide a practical level at which dispersion, coherence, and process-level organization can be evaluated.

For a biological process B containing N_B mapped genes, the patient-level process observable for patient p was defined as

$$O_{B,p} = \{Z_{g,p} : g \in B\},$$

where $Z_{g,p}$ denotes the cohort-wise z -score-normalized expression value of gene g in patient p . The corresponding patient-level BP activity score was defined as

$$\mu_{B,p} = \frac{1}{N_B} \sum_{g \in B} Z_{g,p}.$$

Because $\mu_{B,p}$ is derived from cohort-wise normalized expression values, it represents a signed relative activity score rather than an absolute biological activity measurement. Positive values indicate higher-than-average expression of the mapped BP genes within the cohort, whereas negative values indicate lower-than-average expression.

For the same patient-level BP observable, the distributional width of the mapped gene-expression values was estimated using full width at half maximum:

$$W_{B,p} = FWHM(O_{B,p}).$$

Here, $W_{B,p}$ quantifies the within-process gene-expression width of biological process B in patient p . It is interpreted as an operational measure of observable cloud thickness and as a reproducible proxy for process-level observable dispersion. This interpretation is operational rather than literal: $W_{B,p}$ is not treated as thermodynamic entropy, a direct measure of hidden biological complexity, or the FWHM of BP activity scores across patients. Instead, it describes how broadly the projected BP-level signal is distributed within the mapped gene set for a given patient.

Each patient-level BP observable was therefore represented as an activity–width pair:

$$O_{B,p} \Rightarrow (\mu_{B,p}, W_{B,p}).$$

This representation distinguishes two aspects of the same process-level observation. $\mu_{B,p}$ summarizes the relative activity center of the BP observable, whereas $W_{B,p}$ summarizes its internal distributional width. Thus, two patients or two BP observables may show similar activity scores while differing substantially in observable width, coherence, and interpretability.

For cohort-level visualization and comparison across BP gene sets, patient-level quantities were summarized across all matched patients. The average activity magnitude and average observable width of process B were computed as

$$|\bar{\mu}|_B = \frac{1}{N} \sum_{p=1}^N |\mu_{B,p}|,$$

and

$$\bar{W}_B = \frac{1}{N} \sum_{p=1}^N W_{B,p}.$$

Each BP gene set could therefore be represented in cohort-level activity–width

space as

$$B \Rightarrow (|\bar{\mu}|_B, \bar{W}_B).$$

This cohort-level descriptor summarizes the average relative activity magnitude and the average patient-level observable width of each BP observable. It should be interpreted as a visualization and comparison layer, not as a replacement for patient-level quantities. In particular, $\bar{W}_B$ is the average of patient-level within-BP widths, not the FWHM of BP activity scores across patients.

The analytical workflow is summarized in Figure 1. Bulk gene-expression profiles were mapped onto BP gene sets to compute patient-level BP activity scores, $\mu_{B,p}$, and patient-level observable widths, $W_{B,p}$. These quantities were then summarized across patients to position each BP gene set in cohort-level $|\bar{\mu}|-\bar{W}$ space. Patient-level activity and width values were subsequently used to construct uncertainty-aware discriminability scores for survival-based outcome separation. Detailed transcriptomic preprocessing, BP mapping, z -score normalization, FWHM estimation, survival analysis, and random-validation procedures are provided in the Supplementary Methods.

Together, this representation treats BP-level gene-expression measurements as uncertainty-bearing projected observables rather than direct descriptors of unique biological states. It provides the basis for evaluating observable width, coherence, and discriminability across cancer cohorts.

Figure 1. Conceptual framework of uncertainty-aware discriminability in transcriptomic observables

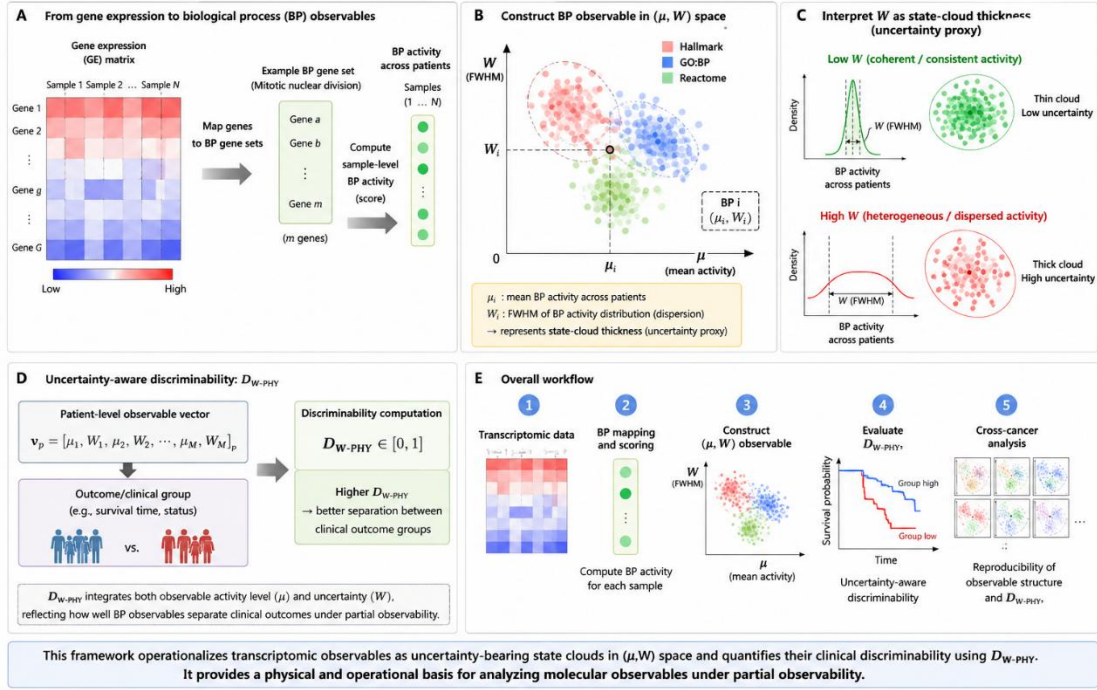


Figure 1. Conceptual framework of uncertainty-aware discriminability in process-level gene-expression observables.

(A) Bulk gene-expression profiles were mapped onto biological-process gene sets to construct patient-level BP activity scores. Each BP observable represents a collective process-level measurement derived from grouped gene-expression signals.

(B) Cohort-level BP observables were represented in joint $|\bar{\mu}|-\bar{W}$ space. Each point represents one BP gene set. $|\bar{\mu}|_B$ denotes the average magnitude of patient-level BP activity scores across the cohort, whereas $\bar{W}_B$ denotes the mean patient-level FWHM-defined width of that BP observable.

(C) Observable width W was interpreted as an operational measure of observable cloud thickness. Low- W observables correspond to compact process-level distributions, whereas high- W observables correspond to broader and more heterogeneous distributions.

(D) Patient-level BP activity scores and patient-level observable widths were combined to compute uncertainty-aware patient-level scores, $D_{W-PHY,p}$, which were then evaluated by survival-based clinical discriminability.

(E) Overall analytical workflow, including transcriptomic preprocessing, BP

mapping and scoring, construction of BP observables in activity–width space, uncertainty-aware discriminability analysis, random-baseline validation, and cross-cancer comparison.

3. Uncertainty-aware discriminability of process-level observables

Conventional pathway-level analyses, including gene set enrichment, pathway activity scoring, and transcriptomic signature methods, usually evaluate biological processes through scalar activity scores [5–8]. These scores remain useful for summarizing process-level activity, but they do not explicitly account for the internal width or compactness of the projected observable. As defined above, two BP observables may show similar activity centers while differing substantially in $W_{B,p}$. This motivates an uncertainty-aware discriminability score that evaluates patient-level process activity together with observable width.

The conceptual distinction between conventional scalar pathway scoring and the proposed activity–width representation is summarized in Table 1. The aim is not to replace pathway activity scores, but to make their interpretation more measurement-aware by asking whether a process-level signal is compact, diffuse, or sensitive to observable-width structure.

Table 1. Conceptual distinction between scalar pathway scoring and the proposed activity–width representation

Aspect	Conventional scalar pathway scoring	Activity–width representation
Basic representation	A pathway or biological process is summarized mainly by one activity score.	A biological-process observable is represented by both activity center and observable width.
Main quantity	Mean, enrichment, or	Patient-level activity $\mu_{B,p}$ and

Aspect	Conventional scalar pathway scoring	Activity-width representation
	pathway activity score.	within-process width $W_{B,p}$.
Treatment of dispersion	Internal dispersion is usually not part of the primary observable.	Dispersion is quantified using $W_{B,p}$, interpreted as observable cloud thickness.
Interpretation	Useful for summarizing process activation or suppression.	Useful for assessing whether the projected process-level signal is compact or diffuse.
Role in this study	Serves as the activity-only baseline.	Serves as the measurement-aware representation used for uncertainty-aware discriminability.

For a selected set of K BP observables, the patient-level activity-magnitude term was defined as

$$M_p = \frac{1}{K} \sum_{B=1}^K |\mu_{B,p}|,$$

where $\mu_{B,p}$ denotes the signed activity score of biological process B in patient p . The absolute value was used because $\mu_{B,p}$ is a relative z -score-based activity measure, and both positive and negative deviations may represent process-level departure from the cohort reference.

To incorporate observable width, we defined the patient-level width-burden term as

$$U_p = \frac{1}{K} \sum_{B=1}^K \log (W_{B,p} + \delta),$$

where $W_{B,p}$ is the FWHM-defined within-process gene-expression width of biological process B in patient p , and δ is a small positive constant used for numerical stabilization. The logarithmic transformation reduces the influence of

extremely broad width values while preserving the contribution of observable-width burden.

The uncertainty-aware patient-level score was then defined as

$$D_{W-PHY,p}(\lambda) = M_p - \lambda U_p,$$

where λ controls the strength of observable-width penalization. The special case

$$D_{W-PHY,p}(0) = M_p$$

corresponds to the activity-only baseline. Thus, $D_{W-PHY,p}(\lambda)$ is not a direct estimate of latent biological state, physical entropy, or patient-specific clinical risk. It is a patient-level stratification score designed to test whether BP activity remains outcome-discriminative after accounting for observable-width burden.

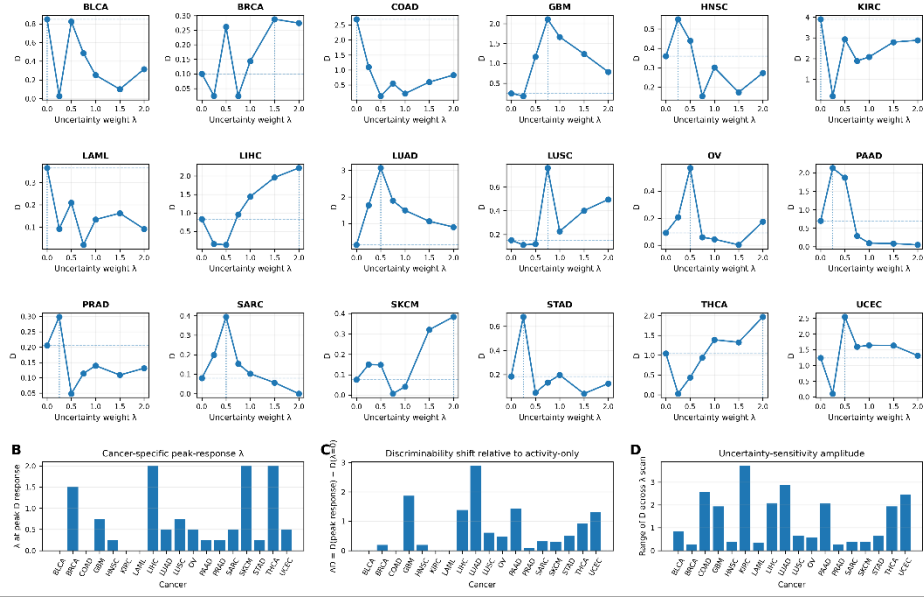
To evaluate clinical discriminability, patients were stratified using $D_{W-PHY,p}(\lambda)$, and survival separation between score-defined groups was quantified using Kaplan–Meier analysis and the log-rank test [39–41]. The survival-based clinical discriminability score was defined as

$$D_{clinical}(\lambda) = -\log_{10}(p_{logrank}[D_{W-PHY,p}(\lambda)]).$$

Larger $D_{clinical}$ values indicate stronger survival separation under the tested score, endpoint, and stratification rule.

We then performed a λ -scan across cancer cohorts to test whether observable width modified survival-based discriminability relative to the activity-only baseline (Figure 2). If activity magnitude alone were sufficient, changing λ would produce little deviation from $D_{clinical}(0)$. Instead, cancer cohorts showed distinct $D_{clinical}(\lambda)$ response profiles, indicating that the contribution of observable width to outcome separation was context dependent. These results support the use of $W_{B,p}$ as an interpretative modifier of discriminability rather than as a universal improvement term. Detailed parameter settings, FWHM estimation, λ -scan procedures, and survival-analysis workflows are provided in the Supplementary Methods.

A



Uncertainty weighting modulates process-level discriminability in a cancer-specific manner, supporting the contribution of cloud-thickness structure beyond activity magnitude alone.

Figure 2. Process-level discriminability exhibits cancer-specific responses to observable-width weighting.

(A) Cancer-specific $D_{clinical}(\lambda)$ response curves across tested uncertainty-weighting values. For each cancer cohort, patients were scored using $D_{W-PHY,p}(\lambda)$ and stratified for survival analysis. Dashed horizontal lines indicate the activity-only baseline ($\lambda = 0$), while vertical dashed lines indicate peak-response weighting regions.

(B) Cancer-specific peak-response uncertainty weighting. Different cohorts exhibited heterogeneous optimal tested λ values, indicating that the contribution of observable width to survival discriminability is context dependent.

(C) Discriminability shift relative to the activity-only baseline, defined as the change in $D_{clinical}$ from $\lambda = 0$ to the best-performing tested λ .

(D) Uncertainty-sensitivity amplitude across cancers, measured as the range of $D_{clinical}$ values observed across the λ -scan. Larger amplitudes indicate stronger dependence of outcome separation on observable-width weighting.

Together, these results show that observable-width weighting modifies survival-based discriminability in a cancer-dependent manner. The purpose of the λ -scan is therefore not to identify a universal optimal λ , but to test how activity-based

outcome separation changes when observable width is included.

4. Real-data organization of process-level observable geometry

We next evaluated whether the activity–width representation provides a practical diagnostic layer for interpreting pathway-level transcriptomic observables. Conventional pathway analyses often rank observables by activity magnitude or outcome association alone. Here, we asked whether adding observable width changes how such signals should be interpreted, particularly in relation to observable coherence, random-background behavior, and observation scale.

For each cancer cohort, Hallmark and biological-process observables were constructed from mapped gene-expression values. Patient-level activity scores, $\mu_{B,p}$, and FWHM-defined observable widths, $W_{B,p}$, were computed for each BP observable and then summarized across patients to generate cohort-level $|\bar{\mu}| - \bar{W}$ landscapes. Detailed preprocessing, gene-set mapping, FWHM estimation, and cohort-level summary procedures are provided in the Supplementary Methods.

Across cancers, BP observables formed non-uniform distributions in activity–width space rather than collapsing into an activity-only pattern (Figure 3A). Processes with comparable average activity magnitude often exhibited different observable widths, indicating that mean activity alone does not fully describe process-level transcriptomic organization. Cancer-by-database summaries further showed that average observable width varied across both cancer contexts and gene-set collections (Figure 3B), suggesting that W provides a comparative axis for evaluating how compact or diffuse BP observables are under different transcriptomic and annotation settings. Coherence-based organization of cancer cohorts provided a descriptive summary of cross-cancer differences in process-level compactness (Figure 3C). This grouping should not be interpreted as supervised cancer classification or mechanistic equivalence, but as evidence

that activity–width coherence profiles contain organizational information. Finally, gene-set-size analysis showed that observable width was scale-sensitive (Figure 3D), indicating that the aggregation scope of each BP observable, N_B , influences the resulting width structure.

Together, Figure 3 shows that W is not merely an auxiliary statistic. It changes the interpretation of pathway-level activity by revealing whether a process-level signal is compact or diffuse. Thus, two BP observables with similar activity magnitude may represent different levels of observable coherence and should not be treated as equivalent.

Figure 3 draft: Cross-cancer organization of activity-width observable geometry

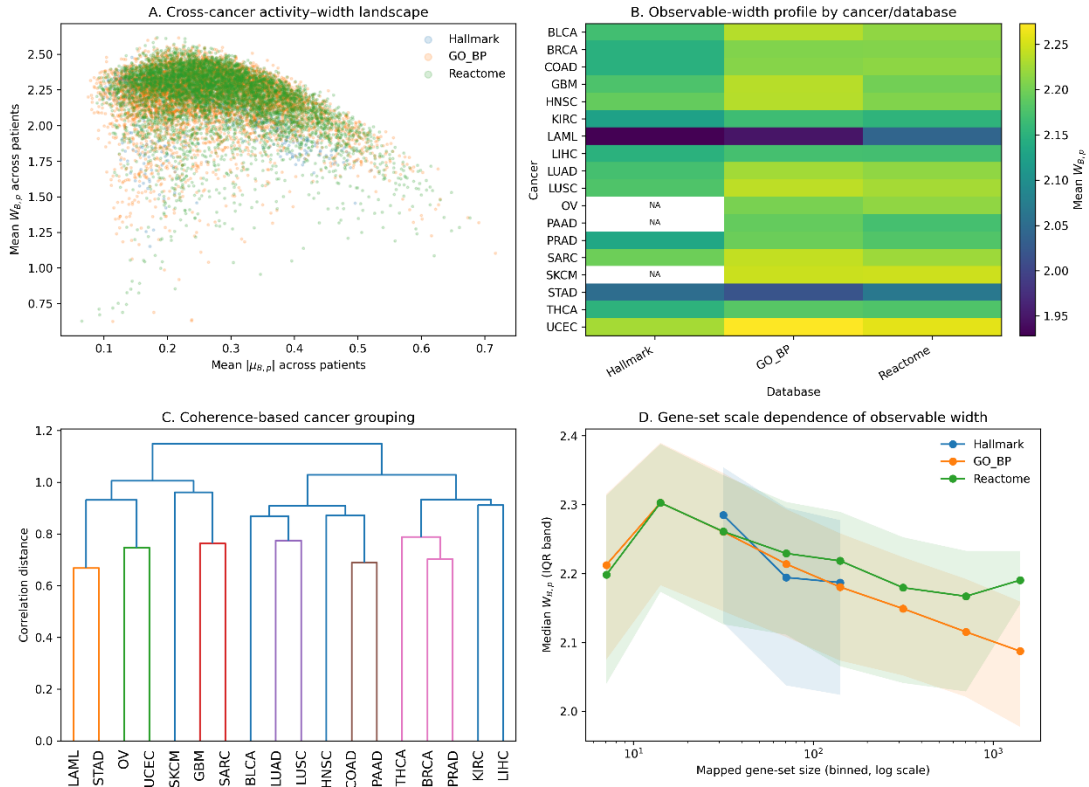


Figure 3. Cross-cancer activity-width geometry of process-level transcriptomic observables.

(A) Cross-cancer activity-width landscape of BP observables. Each point represents one BP gene set summarized across patients by average activity magnitude, $|\bar{\mu}_B|$, and average FWHM-defined observable width, $\bar{W}_B$. The non-uniform distribution indicates that BP observables are not fully described by

scalar activity alone.

(B) Cancer-by-database summary of observable width. Values represent mean patient-level FWHM-defined width summarized across BP observables within each cancer–database combination. NA indicates cancer–database combinations without available Hallmark summary entries in the integrated analysis output.

(C) Coherence-based organization of cancer cohorts using shared BP observable profiles. Coherence reflects the balance between activity magnitude and observable width. The clustering provides a descriptive organization of cancer cohorts by process-level observable compactness, not a supervised cancer classification.

(D) Gene-set scale dependence of observable width. BP observables were grouped by mapped gene-set size, N_B , and median $\bar{W}_B$ values were summarized across size bins. The trend suggests that observable width is scale-sensitive and depends partly on the aggregation scope used to construct process-level observables.

To test whether the observed activity–width organization reflected biologically structured collective behavior rather than random aggregation alone, we compared structured BP observables with size-matched random gene sets. Random baselines were generated independently for each cancer cohort using the same preprocessing, scoring, width-estimation, and survival-discriminability procedures described in the Supplementary Methods. This comparison was not intended to assume that random gene sets are always uninformative. Instead, it was designed to test how discriminability should be interpreted when random projections can also generate high outcome separation.

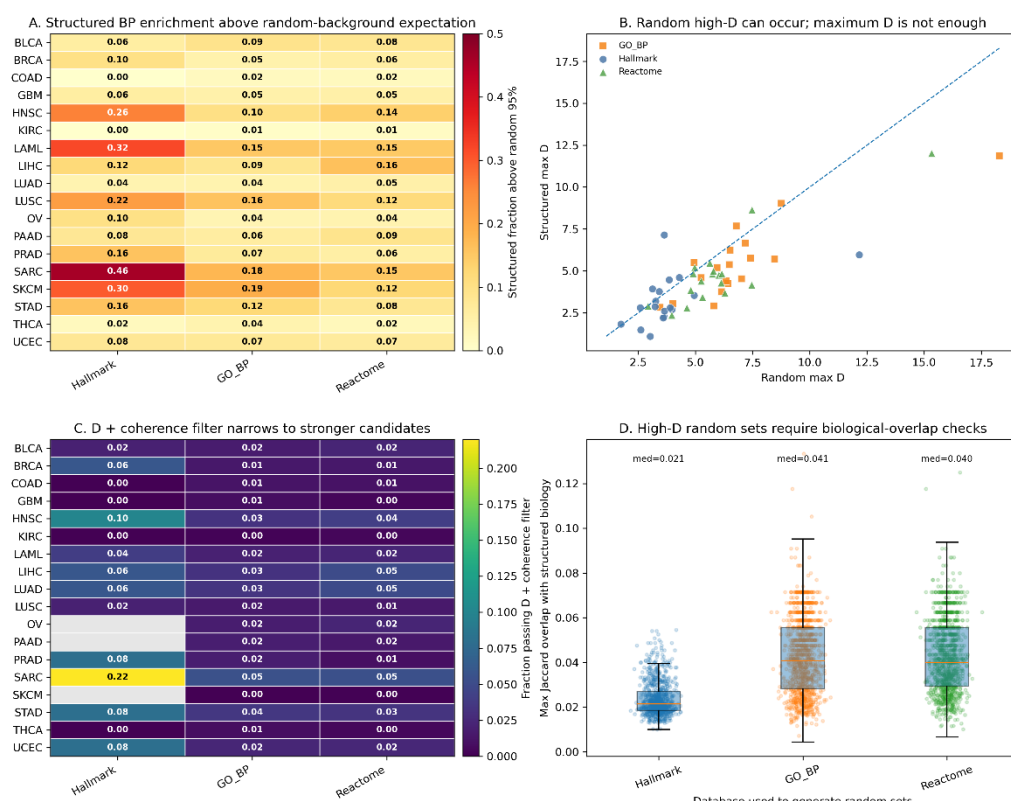
Across cancer cohorts, a subset of structured BP observables exceeded the 95th percentile of size-matched random discriminability distributions, indicating enrichment above random-background expectation in several cancer–database combinations (Figure 4A). However, random gene sets also occasionally reached high maximum discriminability values, sometimes comparable to or higher than

structured BP observables (Figure 4B). This result highlights a key limitation of using discriminability alone as a selection criterion: high D does not automatically imply biological interpretability or reliable process-level organization.

We therefore evaluated discriminability together with width-aware coherence. Structured BP observables that passed both a random-background discriminability threshold and a coherence filter represented a more restricted subset of candidates (Figure 4C). This supports the use of the activity–width framework as a diagnostic tool rather than a simple ranking method. Finally, high-discriminability random sets were examined for overlap with structured biological annotations using maximum Jaccard overlap (Figure 4D). Measurable overlap in some cases suggests that high- D random sets may capture fragments of organized transcriptomic variation, but such signals require additional biological-structure checks before interpretation.

These analyses show that random baselines are not merely negative controls. They distinguish outcome separation alone from more interpretable observables that combine discriminability, width-aware coherence, and biological structure. The proposed framework therefore uses D , W , coherence, random-background comparison, and biological overlap jointly to evaluate whether a high- D signal is likely to reflect structured process-level information or a less interpretable random projection.

Figure 4. Random baselines reveal why D must be interpreted with width-aware coherence and biological structure



Interpretation guide: use random-background expectation (A), recognize that random high-D can occur (B), apply a D + coherence filter (C), and inspect biological overlap of high-D random signals (D).

Figure 4. Random baselines reveal why discriminability must be interpreted with width-aware coherence and biological structure.

(A) Fraction of structured BP observables exceeding the 95th percentile of size-matched random discriminability distributions across cancer cohorts and pathway databases. Higher values indicate stronger enrichment of structured BP observables above random-background expectation.

(B) Comparison between maximum discriminability achieved by structured BP observables and size-matched random gene sets. Random gene sets occasionally achieved high maximum D , indicating that discriminability alone is insufficient for identifying biologically informative observables.

(C) Fraction of structured BP observables passing both a random-background discriminability threshold and a width-aware coherence filter. This panel illustrates how the framework narrows candidate observables by requiring both outcome separation and compact process-level organization.

(D) Biological-overlap analysis of high-discriminability random gene sets. For

each high- D random set, the maximum Jaccard overlap with structured biological annotations was summarized. This analysis evaluates whether random high- D signals overlap with known biological structure or remain less interpretable random projections.

Because BP observables are collective measurements, scale enters the analysis at two levels. At the gene-set level, each BP observable contains a mapped set of genes, N_B , whose aggregation scope can influence observable width, as shown in Figure 3D. At the observable-set level, the patient-level discriminability score is constructed from a selected number of BP observables, denoted by K . We therefore tested whether increasing K improves outcome separation monotonically or instead produces scale-dependent discriminability behavior.

Across cancers, uncertainty-aware discriminability showed non-monotonic dependence on K (Figure 5A–C). Intermediate numbers of selected BP observables often produced stronger discriminability than either very small or very large BP sets. This suggests an observation-lens effect: too few BP observables may incompletely capture process-level organization, whereas too many observables may dilute coherent signals by incorporating broader, weaker, redundant, or less outcome-relevant processes. Therefore, observable quantity alone does not guarantee improved interpretability or clinical separation.

Together, these findings support a scale-aware interpretation of process-level transcriptomic observables. At the gene-set level, observable width depends partly on the aggregation scope of genes within each BP observable. At the observable-set level, discriminability depends on how many BP observables are combined to construct the patient-level score. Thus, the usefulness of a BP representation depends not only on activity and width, but also on whether the observation scope matches the scale of outcome-relevant transcriptomic organization.

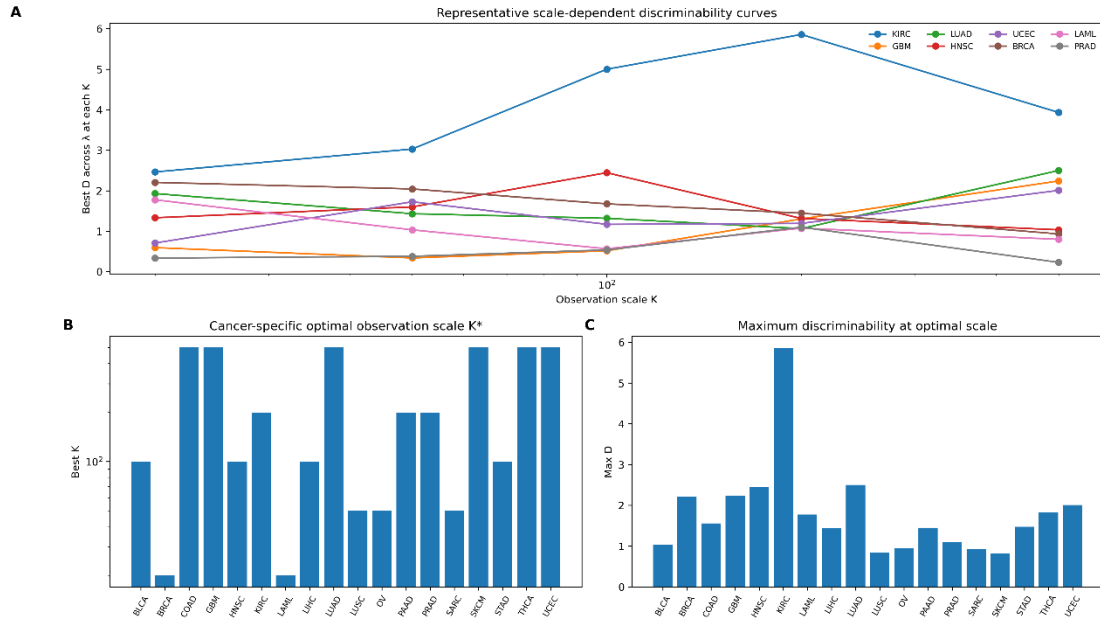


Figure 5. Scale-dependent uncertainty-aware discriminability reveals observation-lens effects across cancers.

(A) Representative scale-dependent discriminability curves across cancers. Uncertainty-aware discriminability varied non-monotonically as the number of selected BP observables, K , increased.

(B) Cancer-specific optimal observation scales, K^* . Different cancer types exhibited distinct numbers of selected BP observables associated with maximal discriminability.

(C) Maximum discriminability achieved at optimal observation scales. Intermediate-scale BP observable sets generally produced stronger discriminability than very small or excessively large observable sets.

Overall, these real-data analyses show that the activity-width representation provides a practical diagnostic framework for pathway-level transcriptomic observables. It does not simply add W to existing pathway scores. Instead, it separates activity magnitude, observable compactness, random-background behavior, biological structure, and observation scale. This provides an operational basis for selecting and interpreting process-level observables under heterogeneous bulk tumor measurements.

5. Discussion

In this study, we introduced a measurement-aware framework for interpreting process-level gene-expression observables in cancer transcriptomes. Rather than treating Hallmark and biological-process observables as scalar activity scores alone, each process-level observable was represented by both its activity center, μ , and its FWHM-defined observable width, W . This activity-width representation provides a simple way to evaluate whether a pathway-level signal is compact, diffuse, or sensitive to projected measurement uncertainty.

The central contribution is not that W universally improves discriminability, but that it changes how discriminability should be interpreted. Conventional pathway-level analyses usually emphasize activity magnitude or outcome association [5–8]. However, two BP observables with similar activity may differ substantially in observable width, coherence, and stability. Likewise, a high-discriminability observable is not automatically biologically meaningful if it lacks coherent activity-width structure or can also be reproduced by high-performing random projections. The proposed framework therefore separates several properties that are often conflated: activity magnitude, observable compactness, survival-based discriminability, random-background behavior, and observation scale.

This distinction is important for bulk tumor transcriptomics. Bulk gene-expression profiles are projected and compositional measurements of heterogeneous tumor tissues rather than direct readouts of unique biological organization [10–17]. Within this setting, W should not be interpreted as literal thermodynamic entropy, a direct measurement of hidden-state complexity, or a decomposition of biological and technical variability. Instead, W is an operational width proxy that captures the dispersion of a process-level transcriptomic signal within the mapped gene set. Such dispersion may arise from cellular mixture, biological fluctuation, gene-set coordination looseness, compensatory regulation, pathway

crosstalk, or measurement variability. Observable geometry therefore provides a measurable trace of uncertainty structure, but not a complete explanation of its biological origin.

The real-data analyses support three practical conclusions. First, BP observables formed measurable activity–width geometry across cancer cohorts. Observable width varied across biological processes, cancer contexts, and gene-set collections, indicating that scalar activity alone does not fully describe process-level transcriptomic organization. Second, random-baseline analyses showed that discriminability must be interpreted cautiously. Random gene sets occasionally achieved high survival separation, meaning that high D alone is insufficient as evidence of biological usefulness. Structured BP observables become more interpretable when survival separation is considered together with width-aware coherence, biological overlap, and reproducibility. Third, discriminability depended on observation scale. At the gene-set level, N_B influenced observable width, whereas at the observable-set level, K influenced patient-level discriminability. These findings support a scale-aware view: adding more genes or more observables does not necessarily improve observability.

The random-baseline results are especially important for interpreting transcriptomic signatures. High- D random sets should neither be dismissed as meaningless noise nor overinterpreted as biological mechanisms. Instead, they should be treated as diagnostic events. A high- D signal becomes more interpretable only when it exceeds random-background expectation, shows coherent activity–width structure, overlaps with known biological annotations, and is reproducible across cohorts or endpoints. This positions the proposed framework as a geometry-aware diagnostic approach rather than a simple pathway-ranking method.

The scale-dependent results further suggest that process-level observation has an effective scope. Very small gene sets may incompletely capture collective

biological organization, whereas overly broad gene sets may average across heterogeneous subprograms and dilute coherent signal structure. Similarly, using too few BP observables may miss relevant process-level variation, while using too many may introduce weak, redundant, broad, or endpoint-irrelevant signals. This has a practical implication for pathway-level analysis: observation design should be scale-aware rather than quantity-driven.

Several limitations should be noted. First, this study focused on bulk gene-expression-derived observables and did not incorporate spatial, temporal, single-cell, proteomic, or perturbational measurements. Second, W is an operational uncertainty-width proxy and does not distinguish biological heterogeneity, projection ambiguity, and technical variability as separate components. Third, the framework does not explicitly model network topology, regulatory hierarchy, or cross-process interactions. These factors may shape observable width and coherence, and future work should test whether pathway topology, co-regulation, protein-interaction structure, or cell-type composition explains variation in W across BP observables. Finally, the present analyses are exploratory and should be validated in independent cohorts and task-specific clinical settings.

Overall, this study provides a measurement-aware interpretation of process-level transcriptomic observables. The activity-width framework does not simply add another pathway score. It changes the basis for interpreting pathway-level signals by separating activity magnitude, observable compactness, random-background behavior, biological structure, and observation scale. These results support evaluating cancer transcriptomic observables not only by scalar activity or survival association, but also by width-aware coherence and scale-dependent discriminability.

6. Conclusion

Bulk gene-expression measurements provide projected observations of

heterogeneous cancer systems rather than direct readouts of unique biological organization. Under this measurement condition, process-level transcriptomic observables may contain uncertainty structure arising from cellular heterogeneity, biological fluctuation, gene-set coordination looseness, and measurement variability [10–17,22–26].

In this study, we introduced a measurement-aware activity–width representation for interpreting process-level gene-expression observables. Instead of representing Hallmark and biological-process observables solely by scalar activity scores, each BP observable was characterized by patient-level activity, $\mu_{B,p}$, and patient-level FWHM-defined observable width, $W_{B,p}$. Here, $W_{B,p}$ serves as an operational measure of observable cloud thickness, not as literal entropy or a direct measurement of hidden biological complexity.

Cross-cancer analyses showed that BP observables exhibit measurable activity–width geometry. Processes with comparable activity magnitude often displayed different observable widths, indicating that scalar activity alone does not fully describe process-level transcriptomic organization. Random-baseline analyses further showed that high discriminability is insufficient by itself, because random gene sets can occasionally achieve elevated outcome separation. Scale-dependent analyses also showed that increasing the number of selected BP observables, K , does not necessarily improve outcome separation.

Overall, this work shows that observable width changes how pathway-level discriminability should be interpreted. The proposed activity–width framework separates activity magnitude, observable compactness, random-background behavior, biological structure, and observation scale. These findings support a measurement-aware approach in which process-level transcriptomic observables are evaluated not only by scalar activity or survival association, but also by observable width, coherence, biological structure, and scale-dependent discriminability [27–33,47–49].

Author Contributions

S.G.K. conceived the study, developed the conceptual framework, performed the analyses, interpreted the results, prepared all figures, and wrote the manuscript.

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Conflict of Interest

The author declares no competing interests.

Data Availability

The datasets analyzed in this study are publicly available from The Cancer Genome Atlas (TCGA) through the UCSC Xena platform:

[UCSC Xena](#)

Detailed cohort information, preprocessing procedures, and pathway resources are described in the Supplementary Methods.

Code Availability

Code used for transcriptomic preprocessing, biological-process mapping, observable-geometry construction, uncertainty-aware discriminability analysis, structured-versus-random validation, and supplementary figure generation is publicly available at: <https://github.com/kensinro/AIDO-D-PHY-I>

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, generative AI tools were used to assist with language refinement and editing. All content was critically reviewed, validated, and revised by the author. The author takes full responsibility for the accuracy, integrity, and originality of the work.

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